Beginning hypothesis testing

Ciaran Evans

Course status so far

So far:

- How can we estimate parameters / fit a model?
  - MLE, method of moments, M-estimation, and applications to regression
- What makes a good estimator?
  - MSE, unbiased estimators, CRLB, consistency, asymptotic relative efficiency
- What is the large-sample behavior of our estimators?
  - WLLN, CLT, delta method, asymptotic normality of the MLE, asymptotic normality of M-estimators

Next: how can we use our estimates for inference?

- Hypothesis tests using Normal distributions
- General hypothesis testing framework
- Other methods for testing hypotheses (Neyman-Pearson, LRT)
- Confidence intervals

Hypothesis tests for a population mean

Let $X_1, X_2, ..., X_n$ be an iid sample from a population with unknown mean μ, and known variance σ^2. We want to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

Test statistic:

Hypothesis tests for a population mean

Let $X_1, X_2, ..., X_n$ be an iid sample from a population with mean μ and variance σ^2. We want to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

Test statistic: $Z_n = \dfrac{\overline{X}_n - \mu_0}{\sigma/\sqrt{n}}$

What if σ is unknown?

Hypothesis tests for a population mean

Let $X_1, X_2, ..., X_n$ be an iid sample from a population with mean μ and variance σ^2. We want to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

Test statistic: $Z_n = \dfrac{\overline{X}_n - \mu_0}{\sigma/\sqrt{n}}$

Rejecting: Should we reject H_0 when Z_n is close to 0, or when Z_n is far away from 0?

Hypothesis tests for a population mean

Let $X_1, X_2, \ldots$ be an iid sample from a population with mean μ and variance σ^2. We want to test

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

Test statistic: $Z_n = \dfrac{\overline{X}_n - \mu_0}{\sigma/\sqrt{n}}$

Rejecting: Reject H_0 when $|Z_n| > z_{\alpha/2}$

p-value: How do we calculate a p-value?

Wald test for one parameter

Wald test for one parameter: examples

- Population mean: $H_0 : \mu = \mu_0$
- Population proportion: $H_0 : p = p_0$
- Regression coefficient: $H_0 : \beta_j = 0$

Testing multiple parameters

Logistic regression model for the dengue data:

$$Y_i \sim Bernoulli(p_i)$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 WBC_i + \beta_2 PLT_i$$

Researchers want to know if there is any relationship between white blood cell count or platelet count, and the probability a patient has dengue.

Question: What hypotheses should they test?

Testing multiple parameters

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.642      0.121  21.772        0
## WBC           -0.289      0.013 -21.533        0
## PLT           -0.007      0.001 -11.061        0
```

$H_0 : \beta_1 = \beta_2 = 0$

Can the researchers test their hypotheses using this output?

Wald tests for multiple parameters

For the dengue example:

$$\widehat{\boldsymbol{\beta}} = \begin{pmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{pmatrix}$$

```
m1 <- glm(Dengue ~ WBC + PLT, data = dengue, family = binom
coef(m1)
```

```
##  (Intercept)          WBC          PLT
##  2.641506279 -0.289290446 -0.006561464
```

```
vcov(m1)
```

```
##                (Intercept)           WBC           PLT
## (Intercept)  1.471934e-02 -4.937020e-04 -5.125888e-05
## WBC         -4.937020e-04  1.804972e-04 -3.221337e-06
## PLT         -5.125888e-05 -3.221337e-06  3.518938e-07
```

Wald tests for multiple parameters

For the dengue example: $\widehat{\boldsymbol{\beta}} = \begin{pmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{pmatrix}$

We want to test: $\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Wald tests for multiple parameters

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \gamma_0$$

For the dengue example:

```
C <- matrix(c(0, 1, 0,
              0, 0, 1), nrow=2, byrow=T)

C
```

```
##      [,1] [,2] [,3]
## [1,]    0    1    0
## [2,]    0    0    1
```

```
C %*% coef(m1)
```

```
##              [,1]
## [1,] -0.289290446
## [2,] -0.006561464
```

Wald tests for multiple parameters

For the dengue example:

```
C
```

```
##      [,1] [,2] [,3]
## [1,]    0    1    0
## [2,]    0    0    1
```

```
vcov(m1)
```

```
##               (Intercept)           WBC           PLT
## (Intercept)  1.471934e-02 -4.937020e-04 -5.125888e-05
## WBC         -4.937020e-04  1.804972e-04 -3.221337e-06
## PLT         -5.125888e-05 -3.221337e-06  3.518938e-07
```

```
C %*% vcov(m1) %*% t(C)
```

```
##               [,1]          [,2]
## [1,]  1.804972e-04 -3.221337e-06
## [2,] -3.221337e-06  3.518938e-07
```

Wald tests for multiple parameters

- $H_0 : \mathbf{C}\beta = \gamma_0$
- Look at $\mathbf{C}\widehat{\beta}$

Test statistic and p-value

- $H_0 : \mathbf{C}\beta = \gamma_0$
- $\mathbf{C}\widehat{\beta} \approx N(\mathbf{C}\beta,\ \mathbf{C}\mathcal{I}^{-1}(\beta)\mathbf{C}^T)$
- Want to turn $\mathbf{C}\widehat{\beta}$ into a scalar test statistic

Example

$H_0 : \mathbf{C}\beta = \gamma_0$

$$(\mathbf{C}\widehat{\beta} - \gamma_0)^T(\mathbf{C}\mathcal{I}^{-1}(\beta)\mathbf{C}^T)^{-1}(\mathbf{C}\widehat{\beta} - \gamma_0) \approx \chi^2_q \qquad \text{under } H_0$$

```r
C %*% coef(m1)
```

```
## [1] -0.289290446 -0.006561464
```

```r
test_stat <- t(C %*% coef(m1)) %*%
  solve(C %*% vcov(m1) %*% t(C)) %*%
  (C %*% coef(m1))
test_stat
```

```
##          [,1]
## [1,] 930.5777
```

```r
pchisq(test_stat, 2, lower.tail=F)
```

```
##               [,1]
## [1,] 8.464858e-203
```